Joint Classification of Hyperspectral and LiDAR Data Using Multiscale Convolution and Cross-Attention-Guided Mamba

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***Abstract*—With the rapid development of deep learning for hyperspectral image (HSI)–light detection and ranging (LiDAR) land-cover classification, Mamba has shown strong potential in feature representation and generalization. However, insufficient semantic interaction among multimodal features and limited multiscale feature modeling still restrict classification performance. To address these challenges, we propose a novel multiscale convolution and cross-attention-guided Mamba (MCCGM) network for joint HSI–LiDAR classification. This framework fully exploits the complementary strengths of distinct architectural components. Specifically, we design a morphological-feature-integrated multiscale CNN (MMC) to extract spatial–spectral features from HSIs, elevation information from LiDAR, and their morphological features, thereby enabling precise characterization of local edges and texture details of ground objects. In parallel, the Mamba fusion encoder (MFE) efficiently models long-range spectral dependencies to promote deeper interaction between HSI spectral features and LiDAR-derived structural information. To further enhance semantic alignment across modalities, a CNN-enhanced cross-attention (CCA) mechanism is embedded into the Mamba block, where multiscale convolutions generate enriched queries, keys, and values to highlight modality-specific correlations. The resulting attention-refined tokens are subsequently processed by Mamba's state-space modeling, which strengthens global dependency learning while preserving intermodality alignment. Extensive experiments conducted on four publicly available benchmark datasets demonstrate that MCCGM consistently outperforms state-of-the-art algorithms, achieving superior classification accuracy and detection performance. The source code is publicly available at https://github.com/cupid6868/MCCGM**

***Index Terms*—Convolutional neural network (CNN), hyperspectral image (HSI), joint classification, light detection and ranging (LiDAR), Mamba.**

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I. Introduction

With the increasing availability of remote sensing satellites, the use of rich remote sensing data to address complex real-world problems has become increasingly feasible and desirable. Remote sensing data now supports a wide range of research areas, such as land-cover classification [1], [2], [3], target detection [4], [5], and saliency detection [6], and has been successfully applied in domains such as medical diagnosis [7], urban planning [8], forest management [9], and mineral exploration [10]. Among these applications, land-cover classification has attracted particular attention and remains one of the most active and widely studied research directions.

In land-cover classification tasks, early works mainly relied on single-source data [11], [12], [13]. Hyperspectral imaging (HSI) is an advanced remote sensing technology capable of capturing detailed spectral measurements across numerous contiguous wavelength bands. By acquiring complete spectral signatures for each pixel and forming a 3-D data cube, HSIs provide significantly richer spectral detail than traditional RGB imagery. As a result, the HSI has demonstrated exceptional performance in land-cover classification. A variety of classical machine learning methods have been explored in early HSI classification research, including support vector machines (SVMs) [14], [15], random forests (RF) [16], k-nearest neighbors (KNNs) [17], and extreme learning machines (ELMs) [18]. With the rapid development of deep learning, more advanced models have been applied to HSI classification, such as convolutional neural networks (CNNs) [19], [20], [21], generative adversarial networks (GANs) [22], [23], recurrent neural networks (RNNs) [24], [25], and Transformers [26], [27], [28], [29]. However, in urban scenarios today, the high similarity of construction materials used in roads and buildings leads to reduced interclass separability, resulting in poor recognition accuracy for certain categories [30]. This issue makes spectral confusion one of the most critical challenges in HSI-based land-cover classification [31]. Recent research has shown that the elevation information provided by LiDAR can mitigate these limitations [32], [33], thereby promoting the development of numerous multimodal joint classification models.

Among these approaches, CNN-based methods have received substantial attention due to their excellent

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performance in image processing, spatial invariance, and feature extraction. Xu et al. [34] proposed a dual-branch CNN architecture in which a dual-channel CNN extracts spatial–spectral features from HSIs and a cascade CNN extracts elevation features from LiDAR. Despite incorporating multimodal information, the fusion strategy employed was suboptimal. Hang et al. [35] extended this line of work by using CNNs to extract diverse features and introducing both feature- and decision-level fusion strategies. However, the contextual richness and feature diversity captured by CNNs remained limited. To address this, Xue et al. [36] and Gao et al. [37] proposed multiscale feature extraction networks, using multiscale CNN modules to capture features across different receptive fields. To further enhance cross-modal information exchange, Fang et al. [38] introduced a cross-modal enhancement module within deep neural networks. Despite the strong classification performance of CNN-based methods, their inherent limitations—including difficulty in capturing long-range spectral dependencies, weak global modeling capability, and inefficiency when handling long spectral sequences—have become increasingly evident.

Transformers excel in handling long sequences and capturing long-term dependencies, effectively overcoming the limitations of CNNs. To address shortcomings in heterogeneous feature representation and multimodal data fusion, Xue et al. [39] employed a spectral-sequence Transformer to process spectral features, a spatial-hierarchy Transformer to extract multilevel spatial features from HSIs and LiDAR, and a cross-attention mechanism for adaptive and dynamic fusion of multimodal data. To further enhance the modeling of complementary multimodal information in remote sensing imagery, Song et al. [40] designed a multisource complementary Transformer. Similarly, Zhang et al. [41] developed the local information interaction Transformer, which performs fine-grained serialization of multisource features and jointly explores HSI and LiDAR characteristics through a local multisource feature interactor. Yang et al. [42] introduced a multimodal data fusion framework built upon stackable modality fusion blocks, which adopt a novel cross-attention mechanism fusion strategy and maintain the integration of spectral and spatial features throughout HSI feature extraction. Zhao et al. [43] combined CNN and Transformer architectures by first extracting shallow spatial–spectral and elevation features using CNNs, followed by deeper feature extraction through a multihead self-attention (SA) Transformer. To enhance feature richness and multimodal interaction, Sun et al. [44] proposed a Transformer with multiscale convolution modules and multimodal information interaction modules. Although Transformer-based methods achieve impressive performance in land-cover classification, their high computational demands and slow inference speeds remain significant challenges.

To address these issues, Mamba has been introduced into classification tasks due to its high computational efficiency and strong capability for long-range dependency modeling [45], [46], [47], [48], [49], [50], [51], [52], [53], [54]. Liao et al. [48] fully considered the edge contour information of images and proposed a joint classification model based on Mamba. Li et al. [47] introduced CMFNet, a cross-Mamba fusion network that improves cross-modal feature fusion, lowers computational redundancy, and performs hierarchical feature extraction using a multilayer design. Zhang et al. [45] proposed an efficient Mamba-based framework for joint HSI–LiDAR classification, which enhances heterogeneous feature extraction and cross-modal fusion through a dual-branch Siamese VSSB, a cross-Mamba feature extraction block, and a modality-guided fusion module.

However, these approaches still exhibit several limitations, which can be summarized in three main aspects.

1) Most existing classification methods rely on relatively simple combinations of CNNs to extract spatial–spectral features within a single-scale context. This often results in the neglect of other important types of information present in remote sensing images, such as morphological characteristics or rich multiscale contextual features.
2) Although Transformer- and Mamba-based methods are effective in modeling long-range dependencies, existing multimodal approaches often conduct sequence modeling on insufficiently aligned modality representations. Consequently, the complementary relationships between HSI spectral information and LiDAR structural cues may not be fully exploited during global dependency learning.
3) Mamba offers an efficient computational framework, particularly well-suited for large-scale image processing, making it a promising candidate for remote sensing applications. Although Mamba has already been introduced into joint HSI–LiDAR classification, existing methods still place relatively limited emphasis on sufficiently explicit cross-modal interaction and modality alignment before global dependency modeling.

To address these limitations, we propose an effective coupling of CNN, attention mechanism, and Mamba, presenting a joint HSI–LiDAR classification model based on multiscale convolution and cross-attention-guided Mamba. Specifically, we first employ convolutional layers with different receptive fields to extract multiscale features and conduct preliminary multilevel fusion. We then introduce a convolution-enhanced multihead cross-attention to facilitate information interaction between HSIs and LiDAR, enabling adaptive and dynamic multimodal fusion. Next, the Mamba unit performs complex transformations on the fused features, generating more discriminative representations and improving model robustness. Finally, an MLP and a classification head complete the classification process. This modeling strategy effectively integrates the strengths of all three frameworks. The main contributions are summarized as follows.

1) We design a morphological feature integrated multiscale CNN (MMC) module that uses convolutional layers with different receptive fields to extract spatial–spectral features from HSIs and elevation features from LiDAR. In addition, morphological convolution is used to capture structural and shape-related information.
2) We introduce a Mamba fusion encoder (MFE) that incorporates a CNN-enhanced cross-attention (CCA) module,

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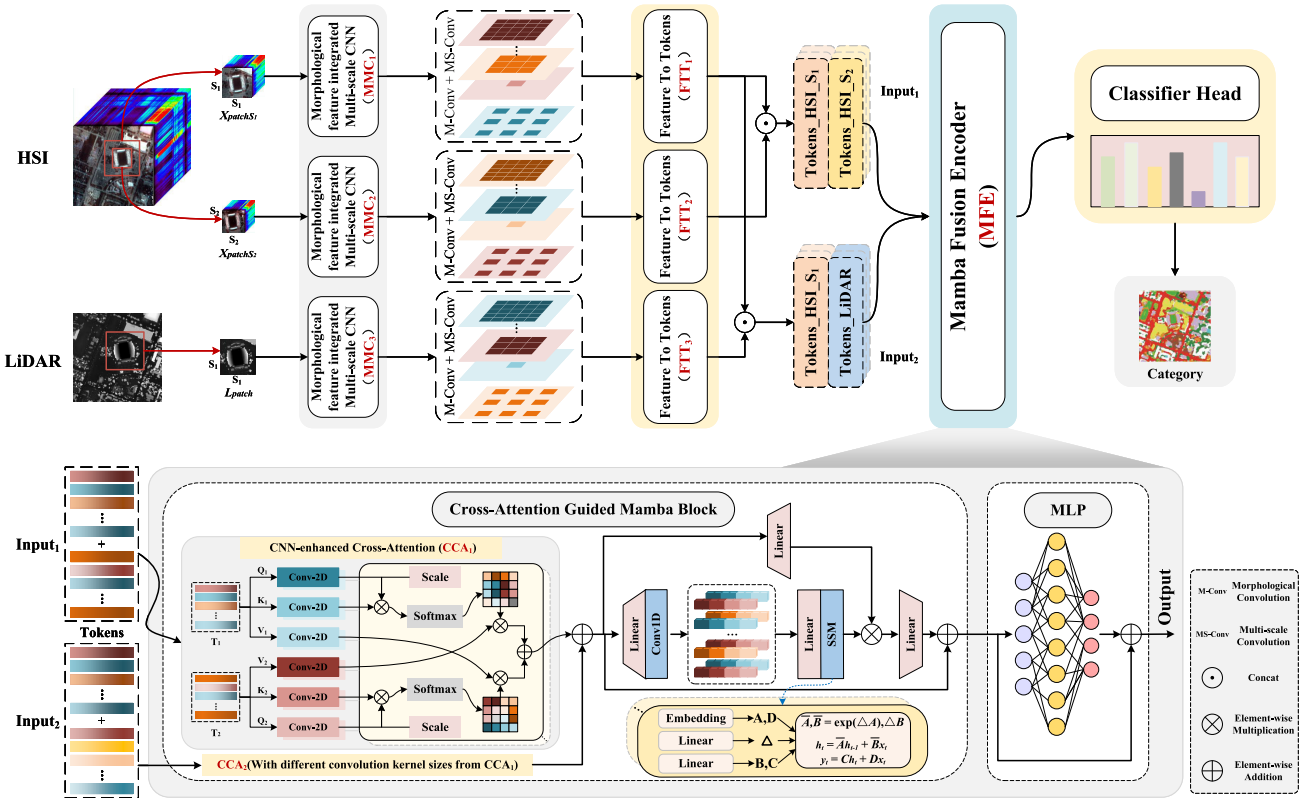


Fig. 1. Overall framework of the proposed MCCGM network. The model takes two HSI patches at different spatial scales and one co-registered LiDAR patch as inputs. The three branches first generate multiscale modality-specific features through MMC, then transform them into token sequences via FTT, and finally perform cross-modal fusion and long-range dependency modeling in the MFE before classification.

enabling effective fusion of multimodal information and improved modeling of long-range dependencies across modalities or spatial regions.

3) In the MFE, the Mamba unit ensures the precise extraction of fine-grained details through an efficient local feature extraction mechanism, guided by the global context captured by the CCA. We validate the effectiveness and generalization capability of the proposed model on four public benchmark datasets.

The remainder of this article is organized as follows. Section II describes the overall architecture of multiscale convolution and cross-attention-guided Mamba (MCCGM) and details its core components. Section III presents the experimental setup, results, and comprehensive discussions. Section IV concludes the article and highlights potential directions for future research.

II. Proposed Method

This section introduces MCCGM, a comprehensive framework for joint classification, comprising five main modules: HSI and LiDAR data preprocessing, MMC, feature to tokens (FTT) module, MFE, and classifier head.

A. Overall Architecture

We propose a novel three-branch joint classification framework that integrates the CNN, attention mechanism, and Mamba architectures. The overall architecture is depicted in Fig. 1. The first and second branches extract shallow spatial–spectral features and morphological features from HSIs using the MMC module at different scales. The third branch extracts shallow elevation and morphological features from LiDAR data covering the same area as the HSI. The FTT module then converts the feature maps into inputs for the cross-attention-guided Mamba architecture. The MFE begins with a CCA mechanism to exchange and fuse feature information from different branches. Subsequently, the Mamba unit performs complex transformations on the fused features to generate more representative features, enhancing model robustness. Finally, the output from the encoder is passed to the classifier head to obtain pixel-level classification results.

B. HSI and LiDAR Data Preprocessing

Typically, we begin with a hyperspectral data cube $\boldsymbol{X} \in \mathbb{R}^{W\times H\times B}$ and LiDAR data $\boldsymbol{L} \in \mathbb{R}^{W\times H\times 1}$ covering the same geographic area, where W and H denote the image width and height, respectively, and B is the number of channels. For each pixel in the HSI, two patch cubes, $\boldsymbol{X}_{patch_{S_1}} \in \mathbb{R}^{S_1\times S_1\times B}$ and $\boldsymbol{X}_{patch_{S_2}} \in \mathbb{R}^{S_2\times S_2\times B}$, are extracted, where the width and height of $\boldsymbol{X}_{patch_{S_1}}$ are S_1, and the width and height of $\boldsymbol{X}_{patch_{S_2}}$ are S_2. Similarly, for each pixel in the LiDAR data, the corresponding patch cube $\boldsymbol{L}_{patch} \in \mathbb{R}^{S_1\times S_1\times 1}$ is extracted. Padding is applied around the image borders to ensure that all patches maintain the required spatial dimensions.

After obtaining the patch cubes for all pixels, those with center pixel labels masked as zero in the ground-truth file

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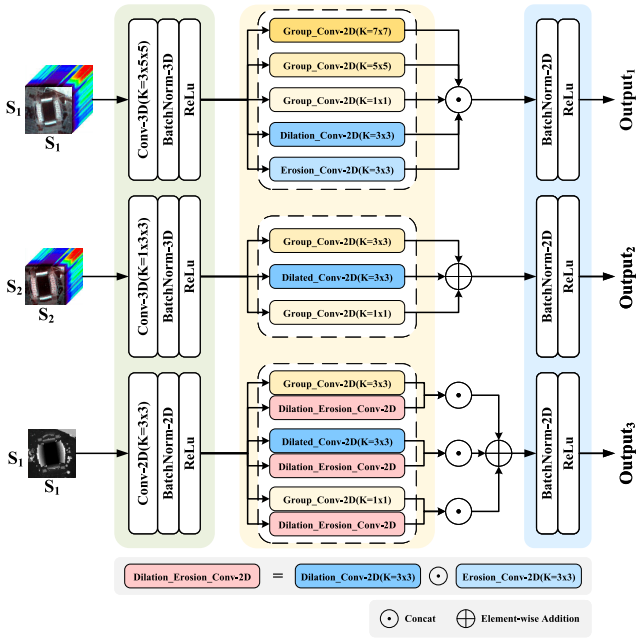


Fig. 2. Detailed illustration of the MMC. The K variable in the figure represents the size of the convolution kernel of the convolution layer.

$\boldsymbol{Y} \in \mathbb{R}^{W\times H\times 1}$ are excluded. The remaining patch cubes are split into training and testing sets.

C. Morphological Feature-Integrated Multiscale CNNs

Based on the powerful feature extraction capabilities of convolution, we design different scale MMC modules with various fusion methods for the three branches, as illustrated in Fig. 2.

Given the rich spectral information in HSI channels, we first apply a 3-D convolution to extract shallow spatial–spectral information from the original image

$$\boldsymbol{F}^{3D}_{MMC_1} = \mathcal{RL}\left(\mathcal{BN}_{3D}\left(Conv3D_{(3\times5\times5)}\left(\boldsymbol{X}_{patch_{S_1}}\right)\right)\right) \tag{1}$$

where $\mathcal{RL}$ represents the rectified linear unit (ReLU) activation function and $\mathcal{BN}$ represents the BatchNorm layer. Different kernel sizes are then used to capture features at various receptive fields. To avoid excessive training parameters, we use grouped convolutions and also incorporate dilated and erosion convolutions for morphological feature extraction [29], [55]. Features are initially fused using concatenation. BatchNorm and ReLU layers are inserted to stabilize training, reduce internal covariate shift, and mitigate gradient vanishing. The specific implementation process can be described by the following formula:

$$\begin{aligned}\boldsymbol{F}_{MMC_1} = \mathcal{RL}\big(\mathcal{BN}_{2D}\big(&Conv2D_{(7\times7)}\left(\boldsymbol{F}^{3D}_{MMC_1}\right)\\ &\odot Conv2D_{(5\times5)}\left(\boldsymbol{F}^{3D}_{MMC_1}\right)\\ &\odot Conv2D_{(1\times1)}\left(\boldsymbol{F}^{3D}_{MMC_1}\right)\\ &\odot DilationConv2D_{(3\times3)}\left(\boldsymbol{F}^{3D}_{MMC_1}\right)\\ &\odot ErosionConv2D_{(3\times3)}\left(\boldsymbol{F}^{3D}_{MMC_1}\right)\big)\big)\end{aligned} \tag{2}$$

where *DilationConv2D* and *ErosionConv2D* represent the dilation convolution layer and erosion convolution layer in morphological convolution, respectively. In detail, (3×3) represents the size of the convolution kernel for this convolutional layer. For the second branch, which has a smaller height and width, we use a 3-D convolution layer with a smaller kernel to capture spatial–spectral features at a smaller receptive field

$$\boldsymbol{F}^{3D}_{MMC_2} = \mathcal{RL}\left(\mathcal{BN}_{3D}\left(Conv3D_{(1\times3\times3)}\left(\boldsymbol{X}_{patch_{S_2}}\right)\right)\right). \tag{3}$$

Subsequent convolutions of varying scales further extract spatial features, combining regular and dilated convolutions to balance computational efficiency and feature extraction capability. Element-wise addition is used for initial feature fusion, with BatchNorm and ReLU layers included. The detailed description of this branch can be summarized as follows:

$$\begin{aligned}\boldsymbol{F}_{MMC_2} = \mathcal{RL}\big(\mathcal{BN}_{2D}\big(&Conv2D_{(3\times3)}\left(\boldsymbol{F}^{3D}_{MMC_2}\right)\\ &\oplus DilatedConv2D_{(3\times3)}\left(\boldsymbol{F}^{3D}_{MMC_2}\right)\\ &\oplus Conv2D_{(1\times1)}\left(\boldsymbol{F}^{3D}_{MMC_2}\right)\big)\big).\end{aligned} \tag{4}$$

The third branch starts with a 2-D convolution layer to initially extract spatial and elevation information from LiDAR data

$$\boldsymbol{F}^{2D}_{MMC_3} = \mathcal{RL}\left(\mathcal{BN}_{2D}\left(Conv2D_{(3\times3)}\left(\boldsymbol{L}_{patch}\right)\right)\right). \tag{5}$$

Three subbranches then further extract features at different receptive fields using grouped and morphological convolutions, with initial feature fusion by concatenation followed by element-wise addition for further fusion. BatchNorm and ReLU layers are also included. The detailed description of the branch can be expressed by the following formula:

$$\begin{aligned}\boldsymbol{F}_{MMC_3} = \mathcal{RL}\big(\mathcal{BN}_{2D}\big(&\big(Conv2D_{(3\times3)}\left(\boldsymbol{F}^{2D}_{MMC_3}\right)\\ &\odot DEConv2D_{(3\times3)}\left(\boldsymbol{F}^{2D}_{MMC_3}\right)\big)\\ &\oplus \big(DilatedConv2D_{(3\times3)}\left(\boldsymbol{F}^{2D}_{MMC_3}\right)\\ &\odot DEConv2D_{(3\times3)}\left(\boldsymbol{F}^{2D}_{MMC_3}\right)\big)\\ &\oplus \big(Conv2D_{(1\times1)}\left(\boldsymbol{F}^{2D}_{MMC_3}\right)\\ &\odot DEConv2D_{(3\times3)}\left(\boldsymbol{F}^{2D}_{MMC_3}\right)\big)\big)\big)\end{aligned} \tag{6}$$

where *DEConv2D* represents two morphological convolution layers, the dilation convolution layer and the erosion convolution layer, connected via the concat method.

D. FTT Module

The SA mechanism in the Transformer excels at processing spectral sequence information and capturing long-range dependencies between pixels and bands. To adapt the framework, we convert the feature maps from the MMC into tokens.

As shown in Fig. 3, we first flatten the feature maps $\boldsymbol{F}_{MMC_1}$, $\boldsymbol{F}_{MMC_2}$, and $\boldsymbol{F}_{MMC_3}$ into vectors. These vectors are then transposed to obtain $\boldsymbol{F}_{flat_1}$, $\boldsymbol{F}_{flat_2}$, and $\boldsymbol{F}_{flat_3}$

$$\left\{\left(\boldsymbol{F}_{flat_{(i)}}\right) = \left(Flatten\left(\boldsymbol{F}_{MMC_{(i)}}\right)\right)^T \mid i = 1,2,3\right\}. \tag{7}$$

The flattened features are multiplied by a learnable weight matrix $\boldsymbol{W}_E$ and transposed, followed by Softmax to obtain $\boldsymbol{F}_{W_{E1}}$, $\boldsymbol{F}_{W_{E2}}$, and $\boldsymbol{F}_{W_{E3}}$

$$\left\{\boldsymbol{F}_{W_{E(i)}} = \left(\boldsymbol{F}_{flat_i}\boldsymbol{W}_{E(i)}\right)^T \mid i = 1,2,3\right\}. \tag{8}$$

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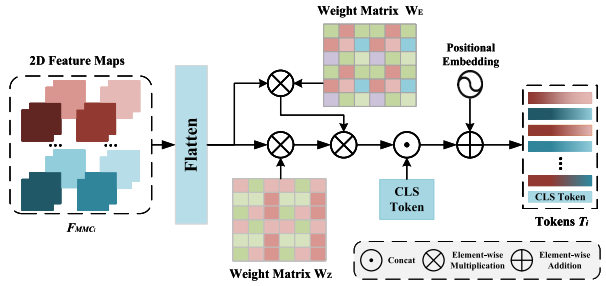


Fig. 3. Visualization of the FTT module. The feature maps from the MMC are flattened and projected by learnable matrices, combined with class tokens and positional embeddings, and then converted into token sequences for the MFE.

Similarly, features are multiplied by another learnable weight matrix $\boldsymbol{W}_Z$ to get $\boldsymbol{F}_{W_{Z1}}$, $\boldsymbol{F}_{W_{Z2}}$, and $\boldsymbol{F}_{W_{Z3}}$

$$\left\{\boldsymbol{F}_{W_{Z(i)}} = \boldsymbol{F}_{flat_i}\boldsymbol{W}_{Z(i)} | i = 1, 2, 3\right\}. \tag{9}$$

These results are then multiplied. Next, a learnable classification token will be added in a concatenated form, while positional information will be embedded, to get the final tokens $\boldsymbol{T}_1$, $\boldsymbol{T}_2$, and $\boldsymbol{T}_3$, with dropout ($\mathcal{DP}$) applied

$$\left\{\boldsymbol{T}_{(i)} = \mathcal{DP}\left(\left(\boldsymbol{F}_{W_{E(i)}}\boldsymbol{F}_{W_{Z(i)}}\right) \oplus \mathcal{PE} \odot \boldsymbol{T}_{cls_{(i)}}\right) | i = 1, 2, 3\right\} \tag{10}$$

where $\mathcal{PE}$ represents positional embedding.

Different from conventional tokenization, the proposed FTT performs a learnable transformation on modality-specific feature maps. Specifically, two learnable matrices are introduced to progressively reorganize the spatial responses and project them into a compact semantic token space. This design enables the generated tokens to preserve more discriminative information while reducing redundancy. Moreover, a CLS token is appended to the token sequence as a learnable global representation carrier, which continuously aggregates informative cues from all tokens during subsequent fusion and sequence modeling. As a result, FTT is more suitable than ordinary tokenization for heterogeneous HSI–LiDAR fusion, where both modality-specific characteristics and global semantic interactions need to be preserved.

E. Mamba Fusion Encoder

As shown in Fig. 1, this encoder is the core module of the proposed model and can be divided into two parts: the cross-attention-guided Mamba block and the multilayer perceptron (MLP). The first part is designed to establish explicit cross-modal interaction before global dependency modeling. Specifically, CCA_1 fuses hyperspectral tokens at different scales to obtain richer global contextual information, while CCA_2 explicitly models the dependency between HSI and LiDAR tokens to enhance cross-modal semantic interaction and modality alignment. Through the collaboration of CCA_1 and CCA_2, the encoder produces a globally fused representation with more complete semantic information and more sufficient multimodal complementarity. Based on these globally interacted representations, the Mamba unit further performs efficient sequence modeling to capture fine-grained local details and propagate informative contextual cues. In this way, the Mamba unit no longer operates on isolated modality-specific tokens, but on semantically enriched and better aligned multimodal features guided by *CCA*. The second part further strengthens the representation capability of the model through nonlinear transformation, feature refinement, and feature projection, thereby improving the global understanding of the input sequence.

Below, we introduce the three key components of this encoder in detail: *CCA*, *Mambaunit*, and *MLP*.

1) CNN-Enhanced Cross-Attention (CCA): As shown in Fig. 1, we use *CCA* with different parameters to process the feature information from different branches. The inputs of the first branch are $\boldsymbol{T}_1$ and $\boldsymbol{T}_2$, while the inputs of the second branch are $\boldsymbol{T}_1$ and $\boldsymbol{T}_3$. During the generation of $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ in multihead SA, 2-D convolution layers with various kernel sizes replace linear layers to enhance the token's feature representation. For example, input $\boldsymbol{T}_1$ and $\boldsymbol{T}_2$ yield $\boldsymbol{Q}_1$, $\boldsymbol{K}_1$, $\boldsymbol{V}_1$, and $\boldsymbol{Q}_2$, $\boldsymbol{K}_2$, $\boldsymbol{V}_2$, respectively. Attention scores are obtained by multiplying $\boldsymbol{Q}_1$ and $\boldsymbol{K}_1$, followed by Softmax and multiplication with $\boldsymbol{V}_2$ to get $\boldsymbol{T}_{atten_1}$

$$\boldsymbol{T}_{atten_1} = Softmax\left(\frac{\boldsymbol{Q}_1\boldsymbol{K}_1^T}{\sqrt{D_1}}\right)\boldsymbol{V}_2 \tag{11}$$

where D_1 represents the dimension of the first input tokens.

Similarly, $\boldsymbol{T}_{atten_2}$ can be derived from $\boldsymbol{Q}_2$, $\boldsymbol{K}_2$, and $\boldsymbol{V}_1$

$$\boldsymbol{T}_{atten_2} = Softmax\left(\frac{\boldsymbol{Q}_2\boldsymbol{K}_2^T}{\sqrt{D_2}}\right)\boldsymbol{V}_1 \tag{12}$$

where D_2 represents the dimension of the second input tokens. Subsequently, $\boldsymbol{T}_{atten_1}$ and $\boldsymbol{T}_{atten_2}$ are preliminarily fused through element-wise addition to obtain the output $\boldsymbol{T}_{First}$ of the first branch

$$\boldsymbol{T}_{First} = \boldsymbol{T}_{atten_1} \oplus \boldsymbol{T}_{atten_2}. \tag{13}$$

Similarly, the output $\boldsymbol{T}_{Second}$ of the second branch is obtained in the same manner. The final output of this component, $\boldsymbol{M}_{atten}$, is obtained by fusing the outputs of the two branches $\boldsymbol{T}_{First}$ and $\boldsymbol{T}_{Second}$ using element-wise addition

$$\boldsymbol{M}_{atten} = \boldsymbol{T}_{First} \oplus \boldsymbol{T}_{Second}. \tag{14}$$

2) Mamba Unit: Beyond cross-attention, which fuses feature information from different branches, the Mamba unit performs complex transformations to generate more representative features and enhance model robustness. The Mamba unit, centered around the state-space model (SSM), describes time-series data statistically. Given input sequence $x(t)$, the SSM maps it to $y(t)$ using hidden state $h(t)$, as per

$$\begin{cases} h'(t) = Ah(t) + Bx(t) \\ y(t) = Ch(t) \end{cases} \tag{15}$$

where A is the state transition matrix and B and C are the mapping matrices. Due to the incompatibility of deep models between continuous systems and discrete sequences, the discrete matrices $\bar{A}$ and $\bar{B}$ can be obtained by the following formula:

$$\begin{cases} \bar{A} = \boldsymbol{exp}(\Delta A) \\ \bar{B} = (\Delta A)^{-1}(\boldsymbol{exp}(\Delta A) - I)\Delta B = \Delta B \end{cases} \tag{16}$$

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where Δ is a scale parameter of time. $\boldsymbol{exp}(\cdot)$ denotes the matrix exponential. I represents the identity matrix. The discretized SSM system can be represented by the following equation:

$$\begin{cases} h_t = \bar{A}h_{t-1} + \bar{B}x_t \\ y_t = Ch_t. \end{cases} \tag{17}$$

We add a convolution to improve efficiency and accelerate the above linear recursive process

$$\begin{cases} \bar{K} = \left(C\bar{B}, C\bar{A}\bar{B}, \ldots, C\bar{A}^{L-1}\bar{B}\right) \\ y = x * \bar{K} \end{cases} \tag{18}$$

where L and K represent the length of the input sequence and the structured convolution kernel, respectively.

Mamba employs selective SSM to effectively capture long-range dependencies. Beyond the SSM mechanism, the Mamba unit also integrates linear transformations, 1-D convolution, and nonlinear activation functions. As shown in Fig. 1, the Mamba unit output $\boldsymbol{M}_{mb}$ with residual connections is obtained as

$$\boldsymbol{M}_{mb_1} = \mathcal{SL}\left(Conv1D\left(Linear\left(\mathcal{LN}\left(\boldsymbol{M}_{atten}\right)\right)\right)\right) \tag{19}$$

$$\boldsymbol{M}_{mb_2} = \mathcal{SL}\left(\boldsymbol{SSM}\left(Linear\left(\boldsymbol{M}_{mb_1}\right)\right)\right) \tag{20}$$

$$\boldsymbol{M}_{mb} = Linear\left(\boldsymbol{M}_{mb_2}\mathcal{SL}\left(Linear\left(\boldsymbol{M}_{atten}\right)\right)\right) + \boldsymbol{M}_{atten} \tag{21}$$

where $\mathcal{SL}$ represents the sigmoid-weighted linear unit (SiLU) activation function and $\mathcal{LN}$ represents the layer normalization layer.

3) Multilayer Perceptron: The MLP performs nonlinear transformations and further extracts and transforms features to enhance the model's representation capability and global understanding of the input sequence. The MLP comprises two linear layers, with Gaussian error linear unit (GELU) activation and dropout layers to capture nonlinear features and prevent overfitting. Similar to the Mamba unit, residual connections are included to mitigate gradient vanishing, facilitating deeper network training and improving performance. The MLP process is represented as

$$\boldsymbol{M}_{mlp} = \mathcal{DP}\left(Linear\left(\mathcal{DP}\left(\mathcal{GL}\left(Linear\left(\boldsymbol{M}_{mb}\right)\right)\right)\right)\right) \tag{22}$$

where $\mathcal{DP}$ represents a dropout layer with a value of 0.1 and $\mathcal{GL}$ represents the GELU activation function.

F. Classifier Head

For final classification, the tokens from the MFE module include a learnable classification token embedded in the FTT module. This token is fed into a simple MLP comprising a LayerNorm layer and a linear layer. The output is a 1-D vector representing the total number of classes in the dataset, with the highest predicted value corresponding to the predicted label for the pixel. The overall execution process is illustrated in Algorithm 1.

Overall, MCCGM is not a simple combination of independent modules, but a unified and task-oriented framework in which different components are tightly coordinated. Specifically, the *MMC* module provides morphology-aware multiscale feature representations, the *FTT* bridges features to token representations, the *CCA* establishes explicit cross-modal interaction and modality alignment, and the *Mambaunit* further performs efficient sequence modeling on the fused representations. Through this collaborative design, MCCGM enables a progressive integration of feature extraction, cross-modal interaction, and global dependency modeling, thereby effectively exploiting complementary HSI–LiDAR information and improving classification performance.

Algorithm 1 MCCGM Model

Input: An HSI data $\boldsymbol{X} \in \mathbb{R}^{W\times H\times B}$, an LiDAR data $\boldsymbol{L} \in \mathbb{R}^{W\times H\times 1}$and a ground-truth data $\boldsymbol{y} \in \mathbb{R}^{W\times H\times 1}$;

1: Set network hyperparameters: The batch size is set to 64, the optimizer is Adam with a learning rate of 5e-3, and the learning rate scheduler is StepLR (with a step size of 50 and a decay factor of 0.9). The number of training epochs ϵ is set to 500.
2: Extract pixel-wise data from the HSI and LiDAR datasets, denoted as $\boldsymbol{X}_{patch_{S_1}}$, $\boldsymbol{X}_{patch_{S_2}}$ and $\boldsymbol{L}_{patch}$, and construct the training and test loaders based on the sampling rate of the training samples.
3: **for** $i = 1$ to ϵ **do**
4: Generate shallow features $\boldsymbol{F}_{MMC_1}$, $\boldsymbol{F}_{MMC_2}$ and $\boldsymbol{F}_{MMC_3}$, using the MMC module by Equations (1)-(6).
5: Introduce learnable classification tokens and positional embeddings through the FTT module, and generate $\boldsymbol{T}_1$, $\boldsymbol{T}_2$, and $\boldsymbol{T}_3$, as described in Equations (7)-(10).
6: Generate the final fused features $\boldsymbol{M}_{mlp}$ using the MFE module by Equations (11)-(22).
7: Extract the learnable classification token and input it into the classifier head to obtain the label corresponding to the current pixel.
8: **end for**
9: The test set is fed into the trained MCCGM model to generate the final predicted classification map.

Output: Predicted labels of the test set.

III. Experiments and Analysis

We conducted experiments on four publicly available datasets to evaluate the effectiveness and robustness of the proposed model. Each dataset contains three files: HSI, LiDAR data, and ground truth. This section provides a detailed description of the experiments. First, we introduce the four datasets used for the experiments. Then, we explain the experimental settings and details, followed by ablation experiments to verify the effectiveness of different modules in the proposed framework.

A. Data Descriptions

We conducted experiments on the following four public datasets: the Houston, Trento, Augsburg, and Muufl Gulfport Hyperspectral and LiDAR Data (MUUFL) datasets.

1) Houston Dataset: Acquired in 2012, this dataset captures scenes from the University of Houston campus and its surrounding urban area. It contains 349 × 1905 pixels, with HSIs having 144 bands. The spatial resolution is 2.5 m, and

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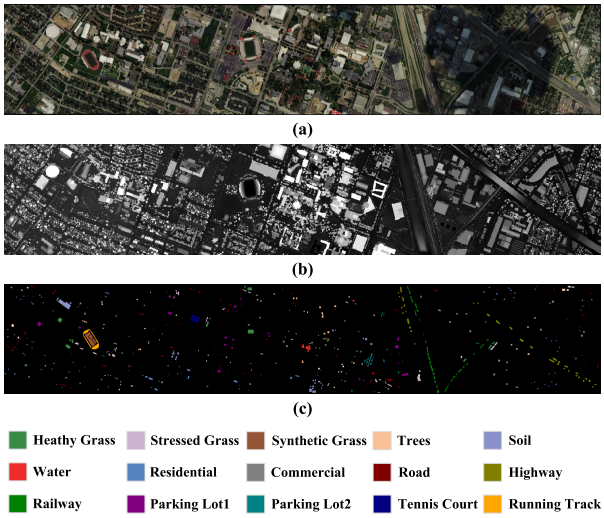


Fig. 4. Houston dataset. (a) Pseudo-color composite image generated by HSIs. (b) Grayscale image generated by LiDAR-based DSM. (c) Ground truth.

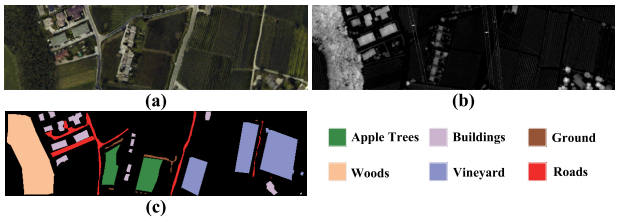


Fig. 5. Trento dataset. (a) Pseudo-color composite image generated by HSIs. (b) Grayscale image generated by LiDAR-based DSM. (c) Ground truth.

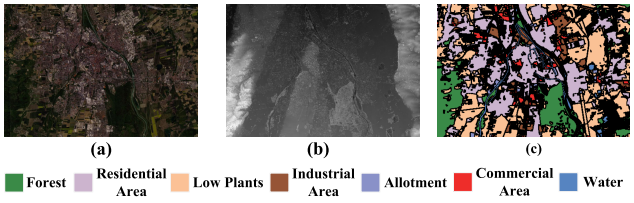


Fig. 6. Augsburg dataset. (a) Pseudo-color composite image generated by HSIs. (b) Grayscale image generated by LiDAR-based DSM. (c) Ground truth.

the dataset includes 15 urban land-cover classes. Fig. 4 shows the visualization of the data in this dataset.

2) Trento Dataset: Captured in the rural areas south of Trento, Italy, this dataset has a spatial dimension of 600×166 pixels. The spatial resolution is 1 m, and the HSI contains 63 spectral bands. The dataset includes six rural land-cover classes. Fig. 5 shows the visualization of the data in the Trento dataset.

3) Augsburg Dataset: Acquired in Augsburg, Germany, this dataset's HSI and LiDAR data are resampled to a uniform spatial resolution of 30 m. It contains 332×485 pixels, with HSIs having 180 bands. The dataset describes seven different land-cover classes. Fig. 6 shows the visualization of the data in the Augsburg dataset.

4) MUUFL Dataset: Acquired in November 2010, this dataset captures scenes from the University of Southern Mississippi's Gulf Park campus. It has a spatial dimension of 325×220 pixels, with HSIs containing 72 spectral bands.

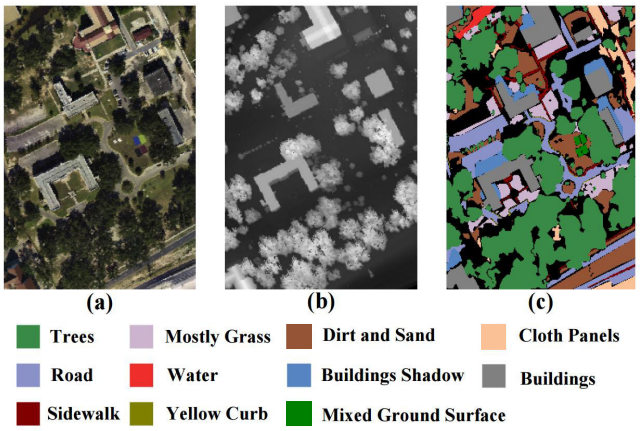


Fig. 7. MUUFL dataset. (a) Pseudo-color composite image generated by HSIs. (b) Grayscale image generated by LiDAR-based DSM. (c) Ground truth.

The dataset includes 11 different land-cover classes. Fig. 7 shows the visualization of the data in the MUUFL dataset.

We present the land-cover class names, the number of training samples, and the number of test samples for these four datasets in Table I.

B. Experimental Settings

1) Evaluation Indicators: We used four commonly used evaluation metrics in the field of land-cover classification: per-class accuracy, overall accuracy (OA), average accuracy (AA), and the kappa coefficient (κ). Higher values indicate better classification performance. Per-class accuracy is the ratio of correctly predicted samples to the total samples in each class. OA is the ratio of correctly predicted samples to the total samples across all classes. AA is the average of the accuracies of all classes. The κ measures the consistency of the model's overall classification results with ground truth.

2) Configurations: To ensure fair comparisons, we conducted experiments using the proposed method and all compared methods on the same hardware server and implemented them using the same PyTorch framework. Our server is equipped with an Intel Xeon Silver 4314 2.40-GHz CPU, 192 GB of RAM, and two NVIDIA GeForce RTX 4090 GPUs with 24 GB of VRAM each. We used Python version 3.8.16 and PyTorch version 2.2.0 for our implementation.

C. Parameter Analysis

We conducted experiments on the following hyperparameters using the controlled variable method to explore the values that allow the proposed model to exhibit the best classification performance.

1) Batch Size: During deep learning training, the batch size primarily affects the training speed and memory utilization. As the batch size increases, the training speed increases but requires higher hardware specifications. Conversely, the smaller batch size slows down the training speed. We chose {8, 32, 64, 128, 256} as a set of candidate values. Fig. 8 shows the experimental results on the four datasets. We found that during training, the best classification performance was achieved when the batch size was 64 (red triangle).

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TABLE I

SAMPLING RATES FOR TRAINING AND TESTING OF DIFFERENT CATEGORIES ACROSS FOUR DIFFERENT DATASETS

NO.	Houston dataset			Trento dataset			Augsburg dataset			MUUFL dataset		
	Class Name	Training.	Test.	Class Name	Training.	Test.	Class Name	Training.	Test.	Class Name	Training.	Test.
C01	Healthy Grass	13	1238	Apple Trees	10	4024	Forest	135	13372	Trees	233	23013
C02	Stressed Grass	13	1241	Buildings	10	2893	Residential Area	303	30026	Mostly Grass	43	4227
C03	Synthetic Grass	7	690	Ground	10	469	Industrial Area	39	3812	Mixed Ground Surface	69	6813
C04	Trees	12	1232	Woods	10	9113	Low Plants	269	26588	Dirt and Sand	18	1808
C05	Soil	12	1230	Vineyard	10	10491	Allotment	6	569	Road	69	6618
C06	Watcr	3	322	Roads	10	3164	Commercial Area	17	1628	Water	5	461
C07	Residential	13	1255				Water	15	1515	Buildings Shadow	22	2211
C08	Commercial	12	1232							Buildings	62	6178
C09	Road	13	1239							Sidewalk	14	1371
C10	Highway	12	1215							Yellow Curb	2	181
C11	Railway	12	1223							Cloth Panels	3	266
C12	Parking Lotl	12	1221									
C13	Parking Lot2	5	464									
C14	Tennis Court	4	424									
C15	Running Track	7	653									
-	Total	150	14879	Total	60	30154	Total	784	77510	Total	540	53147

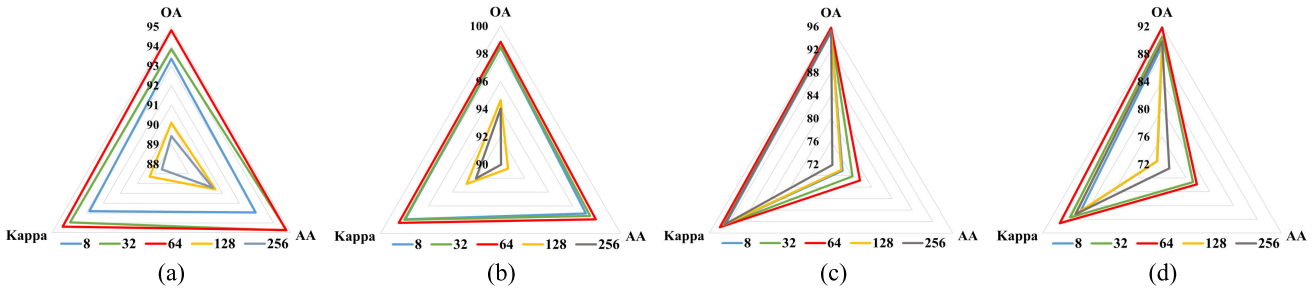


Fig. 8. Impact of different batch sizes on the OA, AA, and κ of the proposed classification model. (a) Houston. (b) Trento. (c) Augsburg. (d) MUUFL.

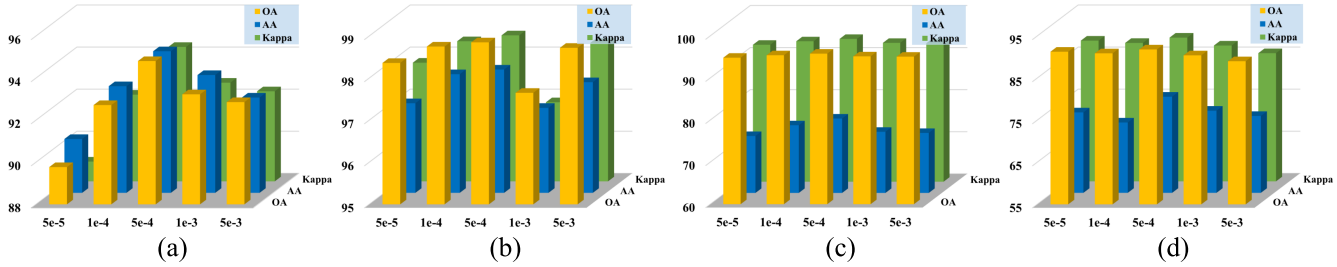


Fig. 9. Impact of different learning rates on the OA, AA, and κ of the proposed classification model. (a) Houston. (b) Trento. (c) Augsburg. (d) MUUFL.

2) Learning Rate: The learning rate is a crucial hyperparameter that directly affects the model's convergence speed. A higher learning rate speeds up convergence but carries the risk of overshooting or divergence. A lower learning rate avoids this risk but slows down training. Based on prior knowledge, we set a group of candidate values {$5e^{-5}$, $1e^{-4}$, $5e^{-4}$, $1e^{-3}$, $5e^{-3}$}. Fig. 9 shows that the best classification performance on the four datasets was achieved when the learning rate was set to $5e^{-4}$.

3) Patch Size for the First Branch: The choice of patch size determines the receptive field of the classification network for the data. Larger patch sizes allow the classification model to capture more contextual information but increase computational complexity and memory pressure. As shown in Fig. 10, we selected a group of candidate values {9, 11, 13, 15, 17}. The best classification performance on all datasets was achieved when the patch size for the first branch was set to 13.

4) Patch Size for the Second Branch: To allow the model to exploit multiscale feature information in the data deeply, we added a branch to process data with different spatial sizes. We selected a group of candidate values {3, 5, 7, 9, 11} for experiments on the four datasets. As shown in Fig. 11, the best classification performance was achieved when the patch size for the second branch was set to 7.

D. Classification Results and Analysis

To verify the classification performance of the proposed model, we compared it with representative classification models, including a linear SA fusion model (LSAF) [56], CoupledCNN [35], a hierarchical CNN and Transformer (HCT) [43], a nearest neighbor-based contrastive learning network (NNCNet) [57], a novel coupled adversarial learning-based classification method (CALC) [58], and a multiscale mixed CNN and multisource data lightweight fusion network (M2FNet) [44]. To demonstrate the advantage of HSIs fused with LiDAR elevation information, we selected single-source classification models that only use HSIs as input, including 3D-CNN [20], multiscale 3-D deep CNN (M3D-CNN) [21], spectral–spatial feature tokenization Transformer (SSFTT) [27], E-Mamba [45], UDM [59],

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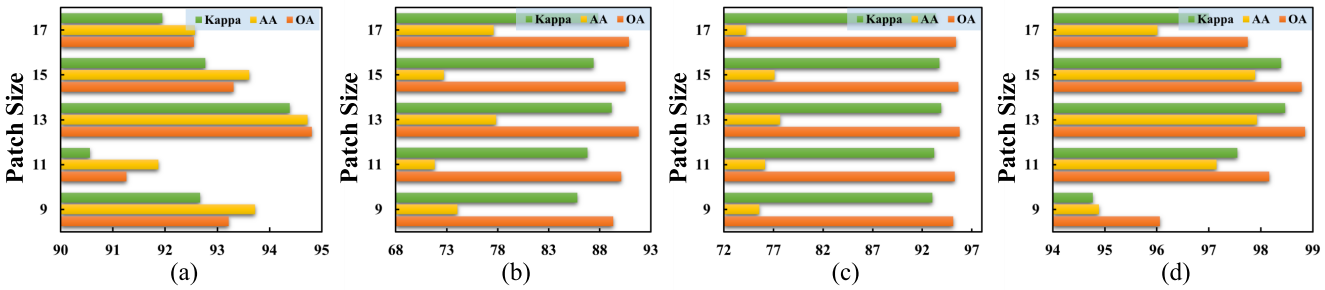


Fig. 10. Impact of different sizes of the first branch patch size on the OA, AA, and κ of the proposed classification model. (a) Houston. (b) Trento. (c) Augsburg. (d) MUUFL.

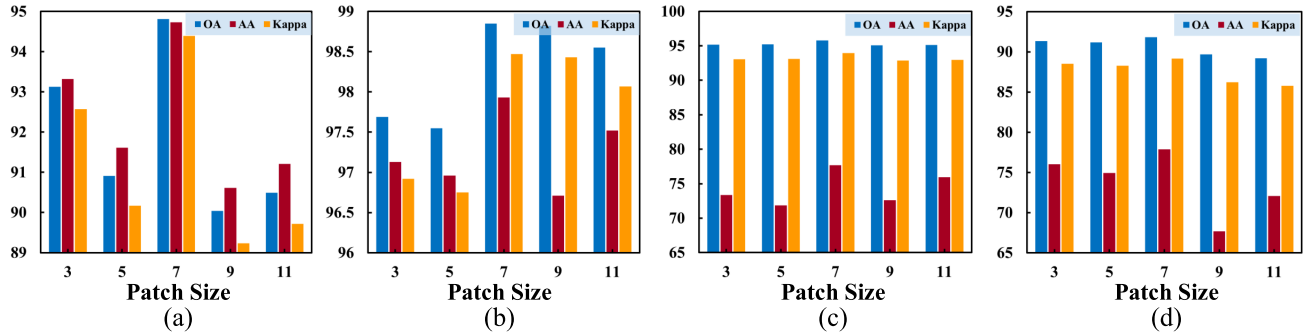


Fig. 11. Impact of different sizes of the second branch patch size on the OA, AA, and κ of the proposed classification model. (a) Houston. (b) Trento. (c) Augsburg. (d) MUUFL.

TABLE II

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE HOUSTON DATASET (OPTIMAL RESULTS ARE BOLDED)

NO.	Only HSI Input						HSI and LiDAR Input							
	3D-CNN	M3D-CNN	SSFTT	morphFormer	SS-Mamba	LSAF	CoupledCNN	HCT	NNCNet	CALC	M2FNet	E-Mamba	UDM	MCCGM
C01	88.29±1.06	88.91±3.18	**95.15±2.38**	89.96±2.63	85.83±9.12	82.93±0.47	83.82±1.58	82.23±1.06	93.94±0.35	91.14±1.45	90.87±5.50	87.85±2.12	91.76±4.72	93.40±1.18
C02	90.36±1.63	94.47±2.22	93.02±3.61	97.53±0.85	94.02±6.41	98.85±0.25	87.13±0.17	99.76±0.42	96.99±0.51	84.72±7.98	93.26±2.89	**99.91±2.12**	98.09±1.49	96.27±1.41
C03	99.37±0.33	98.79±0.30	90.92±6.47	97.97±2.41	99.99±0.05	**100.00±0.00**	97.92±1.21	99.42±0.25	**100.00±0.00**	94.93±2.72	99.23±0.22	**100.00±0.00**	99.17±0.27	99.86±0.14
C04	92.78±2.11	85.69±4.55	98.73±0.63	97.51±0.93	92.98±4.18	74.19±6.72	92.75±3.52	**99.68±0.56**	94.56±0.24	94.59±3.48	93.80±2.36	95.07±1.83	98.35±1.16	98.48±0.83
C05	99.00±0.62	98.10±2.21	99.89±0.19	98.89±1.65	99.92±0.26	99.30±0.68	**100.00±0.00**	**100.00±0.00**	99.95±0.09	99.95±0.09	99.40±0.63	98.31±0.57	**100.00±0.00**	**100.00±0.00**
C06	91.72±1.82	63.87±20.64	78.67±4.41	87.58±0.00	86.68±5.14	83.54±1.89	70.08±4.05	96.38±2.82	86.23±0.90	84.58±7.67	91.30±6.30	**100.00±0.00**	82.67±3.67	85.92±0.65
C07	80.53±0.74	43.96±9.72	83.32±8.48	80.56±3.68	89.82±6.89	78.19±7.30	81.97±1.88	82.44±3.64	87.46±1.09	91.08±6.44	89.06±5.10	86.91±3.79	89.37±5.35	**94.77±2.02**
C08	65.12±2.87	63.31±8.11	64.91±0.82	62.50±5.08	71.61±6.47	74.86±4.81	75.62±3.21	86.31±2.48	81.52±2.82	84.25±1.81	79.14±1.41	**99.89±0.23**	84.41±5.82	92.53±0.45
C09	67.29±8.41	60.10±4.26	70.35±3.35	69.79±2.73	81.16±10.69	71.99±4.23	73.10±4.89	79.66±5.32	84.26±4.84	76.51±3.48	89.48±3.52	**96.56±1.51**	87.43±2.55	88.08±1.10
C10	81.10±2.12	83.37±5.65	85.35±0.72	86.47±3.04	95.57±6.13	92.07±0.78	80.58±4.36	92.62±2.98	88.07±3.30	93.64±7.80	95.34±4.39	90.00±2.29	**96.55±4.87**	95.06±0.62
C11	85.20±3.91	77.21±3.50	87.54±2.51	90.13±1.12	89.48±6.91	97.14±0.93	90.05±2.70	**98.06±1.44**	95.09±1.11	96.40±4.35	97.74±3.57	86.55±5.13	93.95±3.64	97.00±0.50
C12	83.18±3.81	72.56±10.37	91.32±3.74	88.15±1.76	87.18±5.67	79.01±2.59	89.03±0.89	67.92±0.83	86.02±0.34	88.62±4.49	81.60±6.20	76.83±4.63	**92.46±3.14**	89.16±0.99
C13	80.68±2.37	17.53±9.41	**96.84±2.69**	81.90±2.06	53.86±13.55	44.32±17.81	73.99±15.33	78.88±5.65	93.10±0.22	83.26±1.51	89.15±3.16	81.10±2.52	80.45±5.99	90.59±0.69
C14	98.35±0.24	73.58±17.22	**100.00±0.00**	**100.00±0.00**	93.82±6.53	89.39±11.03	**100.00±0.00**	**100.00±0.00**	99.84±0.27	**100.00±0.00**	99.92±0.14	**100.00±0.00**	**100.00±0.00**	99.84±0.27
C15	**100.00±0.00**	99.95±0.09	**100.00±0.00**	**100.00±0.00**	99.77±0.55	98.93±0.85	98.26±3.01	**100.00±0.00**	**100.00±0.00**	99.13±1.50	99.18±1.41	99.69±0.47	99.95±0.07	**100.00±0.00**
OA(%)	85.29±0.64	76.54±1.20	88.20±0.27	87.58±1.04	88.76±0.77	84.99±0.21	86.26±0.47	90.01±0.07	91.85±0.24	90.64±0.54	91.91±0.68	91.84±0.73	93.36±1.05	**94.81±0.34**
AA(%)	86.86±0.56	74.76±2.68	89.07±0.17	88.60±0.84	88.11±0.66	84.31±0.94	86.29±1.12	90.89±0.32	92.47±0.19	90.85±0.81	92.57±0.70	92.04±0.59	92.98±0.93	**94.73±0.28**
$\kappa\times100$	84.09±0.69	74.59±1.32	87.25±0.29	86.57±1.12	87.85±0.84	83.75±0.22	85.14±0.51	89.20±0.07	91.19±0.26	89.88±0.59	91.26±0.73	91.18±0.67	92.82±1.13	**94.39±0.37**

spectral–spatial morphological attention Transformer (MorphFormer) [29], and spectral–spatial Mamba (SS-Mamba) [49].

1) Quantitative Results and Analysis: Tables II–V present the classification metrics of the proposed model and the comparison models on the four datasets, with the highest values for each metric highlighted in bold.

Using the Houston dataset as an example, Table II shows that the proposed model outperformed all other classification models in terms of OA, AA, and κ. Specifically, the proposed model achieved 2.9% higher OA, 2.16% higher AA, and 3.13% higher κ than M2FNet, demonstrating the superior classification performance of the proposed model. Notably, models that used multisource data as input, such as HCT, NNCNet, CALC, M2FNet, and the proposed model, outperformed models that processed single-source data in terms of OA, AA, and κ, highlighting the advantage of HSI fused with LiDAR elevation information for multisource data classification. The proposed model achieved the best classification performance for the classes "soil," "residential," "commercial," and "running track." Although the classification performance for other classes did not always achieve the highest accuracy, the proposed model ranked high for each class. HCT, with its strong spatial–spectral feature extraction capabilities, achieved the best accuracy for the most classes in

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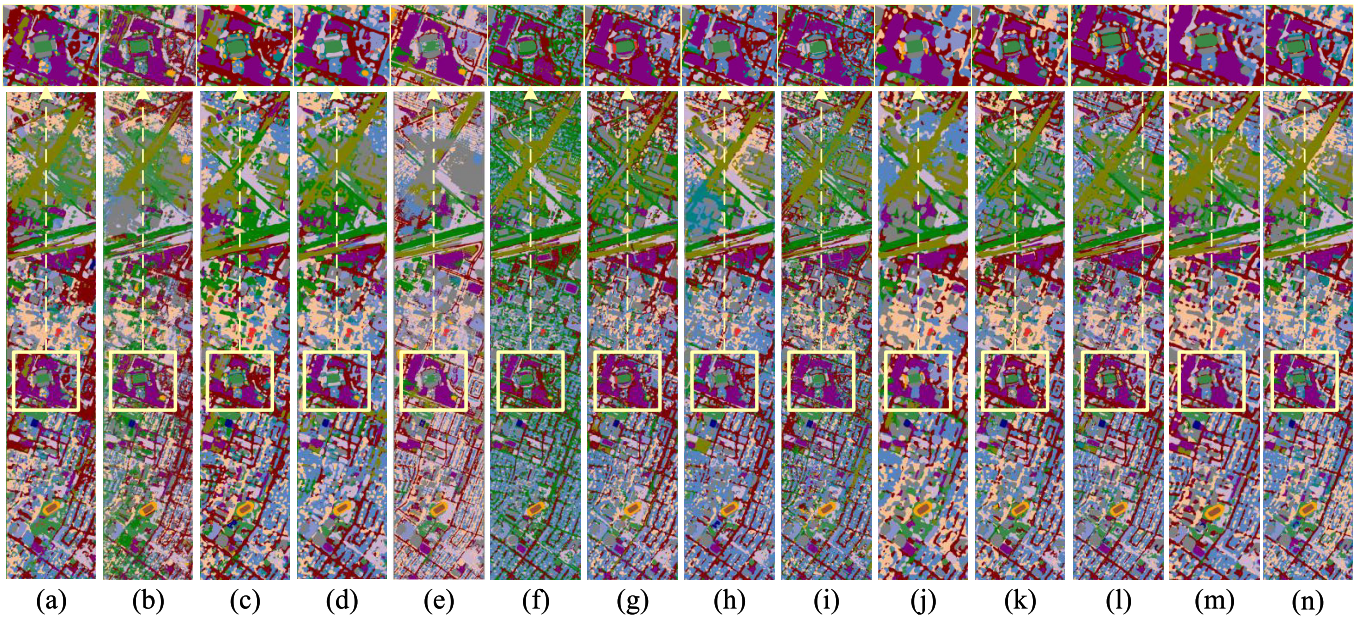


Fig. 12. Visualization classification maps of different methods on the Houston dataset. (a) 3D-CNN. (b) M3D-CNN. (c) SSFTT. (d) morphFormer. (e) SS-Mamba. (f) LSAF. (g) CoupledCNN. (h) HCT. (i) NNCNet. (j) CALC. (k) M2FNet. (l) E-Mamba. (m) UDM. (n) Proposed.

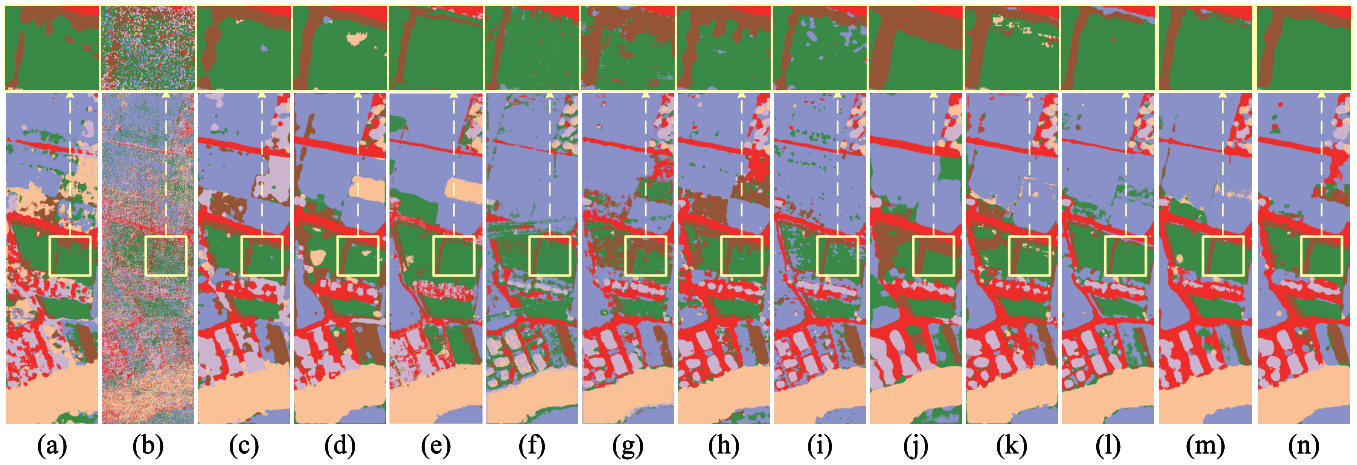


Fig. 13. Visualization classification maps of different methods on the Trento dataset. (a) 3D-CNN. (b) M3D-CNN. (c) SSFTT. (d) morphFormer. (e) SS-Mamba. (f) LSAF. (g) CoupledCNN. (h) HCT. (i) NNCNet. (j) CALC. (k) M2FNet. (l) E-Mamba. (m) UDM. (n) Proposed.

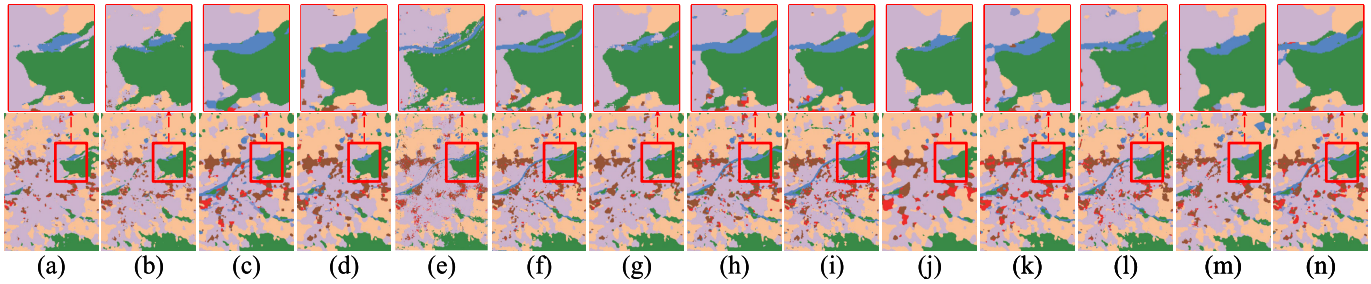


Fig. 14. Visualization classification maps of different methods on the Augsburg dataset. (a) 3D-CNN. (b) M3D-CNN. (c) SSFTT. (d) morphFormer. (e) SS-Mamba. (f) LSAF. (g) CoupledCNN. (h) HCT. (i) NNCNet. (j) CALC. (k) M2FNet. (l) E-Mamba. (m) UDM. (n) Proposed.

terms of single-class accuracy, but its lower accuracy for other classes resulted in significantly lower OA, AA, and κ compared to MCCGM. Additionally, the proposed model demonstrated superior classification performance on the Trento, Augsburg, and MUUFL datasets, achieving 0.54%, 0.91%, and 1.75% higher OA, respectively, than the best method. Thus, the proposed model exhibited strong classification and generalization performance across all four datasets. Meanwhile, owing to the collaborative framework of MCCGM for classification, the standard deviations of the quantitative results are also generally smaller than those of the compared methods, further demonstrating its superior stability.

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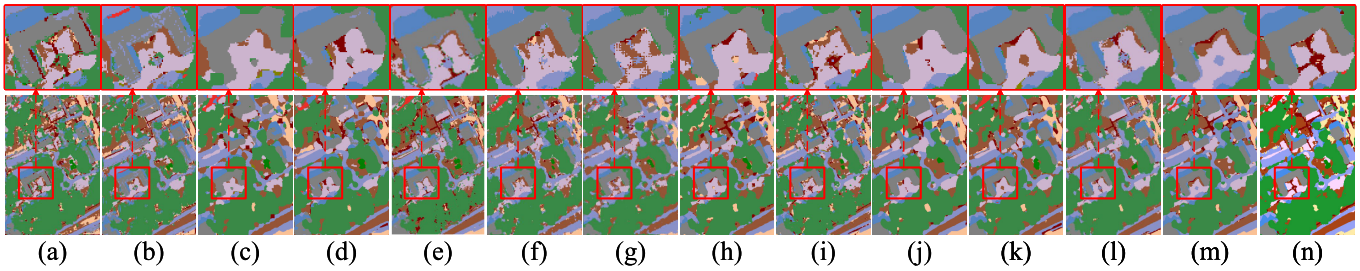


Fig. 15. Visualization classification maps of different methods on the MUUFL dataset. (a) 3D-CNN. (b) M3D-CNN. (c) SSFTT. (d) morphFormer. (e) SS-Mamba. (f) LSAF. (g) CoupledCNN. (h) HCT. (i) NNCNet. (j) CALC. (k) M2FNet. (l) E-Mamba. (m) UDM. (n) Proposed.

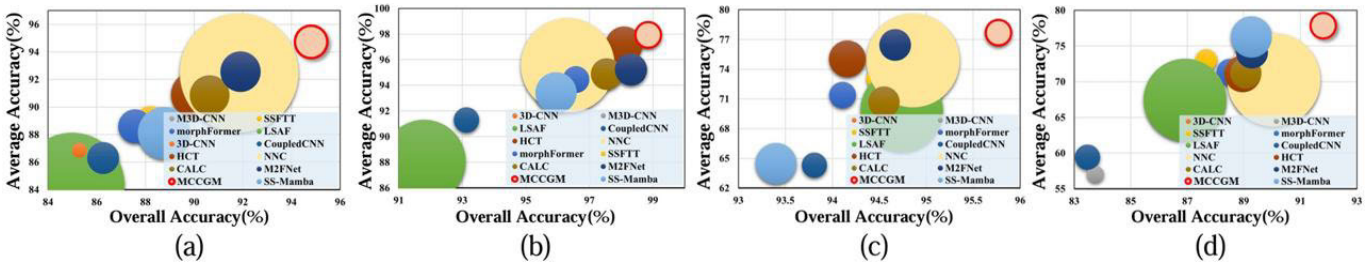


Fig. 16. Comparison of OA, AA, and training parameters of different models. (a) Houston. (b) Trento. (c) Augsburg. (d) MUUFL. The horizontal axis represents the OA metric of the model on a specific dataset, while the vertical axis represents the AA metric of the model on the same dataset. The size of the bubbles indicates the number of training parameters for each model.

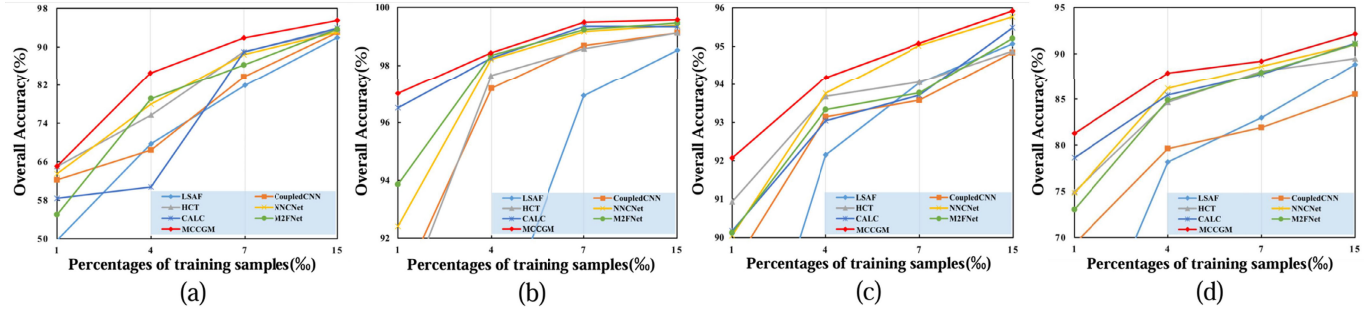


Fig. 17. Effect of different methods on model OA at various sampling rates. (a) Houston. (b) Trento. (c) Augsburg. (d) MUUFL.

2) Visual Evaluation and Analysis: Figs. 12–15 show the visualized classification results of all models on the four datasets. Clearly, the classification maps generated by MCCGM exhibited the least noise, the clearest boundaries, and the closest resemblance to ground truth. To facilitate viewing the visual differences in larger datasets, we selected rectangular windows for magnification. Our model produced the most accurate boundary areas, further demonstrating MCCGM's classification performance.

3) Training Parameter Analysis: In Fig. 16(a)–(d), we compared the training parameters of the proposed model with multisource data classification models. Different colored circles represent different comparison methods, with the coordinates of the circle centers representing each model's OA and AA on different datasets. In other words, the farther the center of the circle is from the origin of the coordinate axes, the better the classification performance of the model it represents. The circle radius indicates the number of training parameters for each model. The proposed model achieved the best classification performance with fewer training parameters.

4) Robustness Evaluation: We conducted experiments on different sampling rates for all methods on the four datasets to verify the robustness of the proposed model. We randomly selected 1‰, 4‰, 7‰, and 15‰ samples for training. Fig. 17(a)–(d) shows that the proposed method maintained the best classification performance on different sampling rates across the datasets. The experimental results indicate that MCCGM consistently maintains strong classification performance under different sampling rates and exhibits smaller performance variations than the compared methods, further demonstrating its robustness and generalization ability. Regarding the data split strategy, Table I presents the detailed training and testing partitions for all datasets. As shown in the table, the number of training samples is intentionally kept relatively small to better reflect the practical challenge of limited labeled data in HSI–LiDAR joint classification. Meanwhile, reserving the majority of labeled samples for testing allows model performance to be evaluated on a sufficiently large test set, thereby improving the reliability and statistical significance of the reported results. Such a split protocol is also suitable for assessing the representation and generalization capability of the model under limited-label conditions. For the Trento dataset, we adopt a fixed setting of 10 training samples per class, which is mainly determined by the class

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TABLE III

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE TRENTO DATASET (OPTIMAL RESULTS ARE BOLDED)

NO.	Only HSI Input						HSI and LiDAR Input							
	3D-CNN	M3D-CNN	SSFTT	morphFormer	SS-Mamba	LSAF	CoupledCNN	HCT	NNCNet	CALC	M2FNet	E-Mamba	UDM	MCCGM
C01	97.19±1.51	54.22±19.55	97.54±0.52	97.93±0.25	**99.96±0.05**	95.44±2.77	82.15±14.86	99.24±0.11	94.36±0.37	97.85±1.71	99.30±0.53	99.64±0.41	99.29±0.50	98.50±0.71
C02	68.41±4.93	44.07±12.46	92.53±2.22	82.26±0.69	90.68±10.97	79.42±4.30	85.47±10.86	97.71±1.95	**99.26±0.04**	98.99±1.76	97.34±1.27	95.92±0.53	93.68±3.20	97.41±0.33
C03	97.30±0.98	64.32±11.96	97.51±0.33	99.29±1.23	98.40±0.75	88.06±3.10	97.23±2.10	97.80±0.12	96.52±0.33	89.84±5.42	84.79±4.23	**100.00±0.00**	87.66±5.04	97.87±0.00
C04	99.60±0.42	85.71±07.65	99.97±0.05	99.74±0.22	99.99±0.02	92.17±5.34	99.53±0.51	99.99±0.03	99.97±0.01	**100.00±0.00**	99.88±0.09	99.99±0.03	**100.00±0.00**	**100.00±0.00**
C05	82.95±4.25	41.33±17.34	98.20±1.04	99.83±0.17	99.46±0.77	98.87±0.93	95.19±2.92	98.73±1.42	95.74±0.36	99.61±0.63	99.89±0.06	99.11±0.57	**100.00±0.00**	99.90±0.10
C06	72.39±0.70	38.85±03.00	75.48±2.67	87.64±1.12	72.00±6.88	74.52±7.90	88.27±0.91	89.46±3.51	87.46±0.08	83.00±4.66	90.20±1.37	**98.78±1.17**	98.03±1.25	93.91±0.87
OA(%)	87.60±2.18	56.82±5.81	95.71±0.33	96.58±0.18	95.95±0.06	91.80±1.23	93.13±1.82	98.09±0.35	96.32±0.07	97.54±0.76	98.31±0.08	**99.11±0.21**	98.97±0.04	98.85±0.05
AA(%)	86.31±1.81	54.75±4.55	93.54±0.65	94.45±0.40	93.42±0.67	88.08±0.30	91.31±3.30	97.15±0.23	95.55±0.04	94.88±1.22	95.23±0.68	**98.03±0.37**	96.44±0.55	97.93±0.11
$\kappa\times100$	83.61±2.85	45.15±6.78	94.28±0.43	95.43±0.24	94.60±0.09	89.07±1.58	90.88±2.44	97.46±0.46	95.09±0.09	96.72±1.01	97.75±0.11	**98.81±0.26**	98.61±0.05	98.47±0.07

TABLE IV

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE AUGSBURG DATASET (OPTIMAL RESULTS ARE BOLDED)

NO.	Only HSI Input						HSI and LiDAR Input							
	3D-CNN	M3D-CNN	SSFTT	morphFormer	SS-Mamba	LSAF	CoupledCNN	HCT	NNCNet	CALC	M2FNet	E-Mamba	UDM	MCCGM
C01	98.12±0.02	98.51±0.42	97.49±0.70	98.09±0.59	98.86±0.01	99.59±0.12	98.44±0.33	98.36±0.03	98.84±0.47	96.75±1.03	98.20±0.11	97.34±0.12	**99.22±0.45**	98.75±0.07
C02	**98.80±0.13**	98.12±0.24	98.29±0.12	97.11±0.35	97.81±0.31	98.41±0.17	97.81±0.11	97.21±0.53	98.19±0.38	98.46±0.36	97.82±0.47	95.78±0.31	98.37±0.28	98.65±0.18
C03	56.64±3.03	58.54±6.01	76.19±1.11	84.31±3.01	68.44±5.90	77.26±2.41	77.85±4.00	77.20±4.05	76.54±2.51	84.27±1.95	84.16±1.75	76.73±3.19	81.32±2.55	**86.11±6.51**
C04	95.62±0.32	98.04±0.45	97.36±0.05	97.19±0.26	97.75±0.38	98.49±0.16	98.36±0.08	97.49±0.64	98.57±0.14	97.78±0.16	97.15±0.54	98.70±0.23	98.37±0.17	**98.75±0.16**
C05	00.82±0.81	00.35±0.35	33.80±1.37	30.76±4.66	17.22±4.23	35.15±6.24	14.47±9.32	64.73±16.26	67.14±4.96	27.01±9.25	63.21±2.46	**94.66±1.06**	34.58±13.42	65.61±2.39
C06	13.78±1.44	12.16±4.93	52.64±3.62	40.29±1.17	26.11±3.21	22.97±3.02	17.69±2.31	37.16±7.43	32.13±4.58	41.13±4.48	39.19±10.68	**53.28±2.92**	26.83±3.63	40.27±17.61
C07	37.07±1.16	34.85±1.15	55.73±1.86	52.01±2.06	45.18±2.01	56.77±3.03	45.52±1.19	53.20±2.22	52.78±1.07	49.37±7.49	55.42±1.28	**80.86±0.82**	48.63±7.13	55.60±2.96
OA(%)	91.80±0.27	92.45±0.21	94.48±0.11	94.11±0.07	93.40±0.10	94.74±0.24	93.81±0.09	94.16±0.23	94.86±0.07	94.55±0.02	94.67±0.17	95.36±0.15	94.46±0.29	**95.77±0.04**
AA(%)	57.26±0.73	57.23±0.66	73.07±0.88	71.39±0.62	64.48±0.61	69.81±2.03	64.31±0.94	75.05±1.80	74.88±0.65	70.68±1.51	76.45±0.91	76.81±0.59	69.62±2.40	**77.68±1.67**
$\kappa\times100$	87.99±0.42	88.98±0.32	92.06±0.17	91.54±0.10	90.48±0.13	92.40±0.36	91.06±0.14	91.61±0.33	92.60±0.10	92.12±0.04	92.35±0.24	93.32±0.17	91.89±0.43	**93.93±0.05**

TABLE V

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE MUUFL DATASET (OPTIMAL RESULTS ARE BOLDED)

NO.	Only HSI Input						HSI and LiDAR Input							
	3D-CNN	M3D-CNN	SSFTT	morphFormer	SS-Mamba	LSAF	CoupledCNN	HCT	NNCNet	CALC	M2FNet	E-Mamba	UDM	MCCGM
C01	95.71±0.22	96.29±0.12	94.60±0.92	95.47±0.15	96.12±0.83	96.90±0.16	93.12±0.68	96.67±0.34	96.75±0.55	96.37±0.46	97.18±0.46	96.57±0.43	**98.21±0.14**	97.60±0.38
C02	72.52±1.22	68.19±2.41	79.28±1.14	80.15±2.13	79.35±5.86	69.30±3.84	73.24±4.85	83.23±0.65	80.59±1.66	81.89±3.64	77.12±2.50	73.20±4.19	**88.84±2.66**	80.88±3.18
C03	70.34±0.85	75.87±2.07	79.96±3.56	83.72±2.56	82.35±6.56	79.07±1.36	78.02±1.77	79.28±0.71	82.03±2.62	78.72±1.27	82.88±1.33	82.47±0.81	**87.98±1.86**	84.26±0.32
C04	78.61±1.17	68.40±4.52	85.29±1.98	84.83±6.41	88.79±3.80	79.68±0.95	71.52±3.88	77.49±3.66	85.48±0.20	84.61±2.94	**90.74±2.47**	89.81±0.29	66.74±15.33	89.27±0.77
C05	87.85±0.52	84.90±3.93	91.24±1.75	90.15±2.07	89.98±2.13	92.88±1.18	85.86±1.60	91.90±0.16	94.03±0.32	92.97±1.81	89.56±0.53	87.70±1.35	**94.84±0.19**	94.48±0.52
C06	34.63±8.01	22.85±7.20	**95.23±2.50**	78.89±10.95	83.11±8.84	95.16±1.23	58.28±4.13	91.97±2.07	86.12±2.15	69.92±23.43	85.03±7.04	95.02±1.84	42.41±15.27	94.50±3.16
C07	59.01±9.80	63.03±5.80	71.60±6.81	75.59±3.13	82.58±4.40	64.09±5.38	48.88±4.34	78.46±5.32	82.04±1.34	77.88±4.91	79.38±2.01	72.77±3.41	83.46±8.71	**87.86±1.07**
C08	72.03±1.64	88.45±5.17	94.44±1.49	93.92±0.54	91.01±2.60	92.74±1.42	94.56±0.40	95.26±1.29	96.33±0.84	**96.90±0.79**	91.43±0.77	95.96±0.36	93.50±1.82	96.85±0.57
C09	46.92±2.55	29.66±3.50	27.13±1.84	33.94±2.81	53.72±7.18	19.62±2.51	20.11±3.43	32.09±3.68	44.71±1.24	31.12±9.38	45.59±3.40	**62.93±2.86**	44.37±10.83	54.97±7.94
C10	13.81±3.87	06.81±2.61	11.60±3.36	04.05±0.84	**14.53±6.91**	03.87±0.96	00.55±0.55	01.66±0.55	06.35±0.39	00.00±0.00	04.79±0.64	00.00±0.00	00.40±0.57	08.47±2.23
C11	56.14±0.78	22.93±4.43	73.31±10.12	64.16±13.51	77.44±13.38	48.37±12.88	29.57±3.86	54.14±15.0	18.42±3.19	73.18±7.73	71.05±4.89	**100.00±0.00**	47.62±25.12	67.17±2.14
OA(%)	82.51±0.19	83.75±1.63	87.67±0.14	88.51±0.05	89.27±0.72	86.92±0.70	83.47±0.34	88.96±0.15	90.06±0.02	89.07±0.35	89.29±0.05	89.65±0.12	91.48±0.38	**91.81±0.15**
AA(%)	62.51±0.54	57.04±1.59	73.06±1.32	71.35±1.75	76.27±1.79	67.43±2.27	59.43±0.88	71.10±0.76	70.26±0.73	71.23±2.30	74.07±0.77	64.11±1.82	68.03±0.54	**77.85±1.00**
$\kappa\times100$	76.69±0.32	78.27±2.21	83.67±0.15	84.82±0.07	85.82±0.93	82.46±0.98	78.04±0.43	85.29±0.18	86.82±0.07	85.51±0.45	85.76±0.04	86.30±0.19	88.46±0.55	**89.16±0.20**

distribution and sample-scale characteristics of this dataset. If the same 1.5% sampling rate were directly applied, some classes would contain too few training samples, leading to an imbalanced training distribution and consequently affecting training stability and classification performance, as illustrated in Fig. 16.

5) Efficiency Evaluation: Table VI compares the computational efficiency of different methods in terms of FLOPs, parameter count, and runtime. It can be seen that MCCGM is at a relatively moderate level in overall computational complexity. Specifically, although its FLOPs and parameter count are higher than those of lightweight CNN methods such as CoupleCNN and LSAF, they are significantly lower than models with high computational costs, such as 3D-CNN and M3D-CNN. In terms of runtime, the inference time of MCCGM is higher than most CNN and Transformer methods, mainly due to the additional computational overhead introduced by the three-branch structure and extra components.

TABLE VI

COMPARISON OF THE EFFICIENCY OF DIFFERENT METHODS

Model	FLops/M	Params/K	Running time/ms
3D-CNN	82.43	798.90	0.424
M3D-CNN	12.41	1462.37	0.432
CoupleCNN	2.99	38.41	0.419
SSFTT	8.63	51.40	1.489
LSAF	5.92	164.29	0.416
HCT	21.93	545.73	1.101
E-Mamba	8.90	125.50	7.503
UDM	25.91	369.65	7.551
MCCGM	14.48	186.84	5.640

However, compared with E-Mamba and UDM, MCCGM has higher computational efficiency. Overall, MCCGM achieves a good balance between computational efficiency and classification performance.

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TABLE VII
ABLATION ANALYSIS OF DIFFERENT MODAL DATA INPUTS (OPTIMAL RESULTS ARE BOLDED)

Cases	Houston			Trento			Augsburg			MUUFL		
	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$
Only HSI	90.34	91.79	89.56	94.57	92.96	92.79	94.82	75.99	92.57	89.69	74.14	86.34
Only LiDAR	58.67	56.87	55.30	86.91	79.49	82.46	85.20	49.84	78.39	71.93	45.51	62.15
HSI+LiDAR	**94.81**	**94.73**	**94.39**	**98.85**	**97.93**	**98.47**	**95.77**	**77.68**	**93.93**	**91.81**	**77.85**	**89.16**

TABLE VIII
ABLATION ANALYSIS OF DIFFERENT COMPONENTS IN THE PROPOSED MODEL ON THE HOUSTON DATASET (OPTIMAL RESULTS ARE BOLDED)

Cases	Component					Indicators		
	MMC	FTT	CCA	Mamba unit	MLP	OA (%)	AA (%)	$\kappa \times 100$
01	✓	✗	✗	✗	✗	76.39	73.87	74.44
02	✓	✓	✗	✗	✗	90.22	90.01	89.42
03	✓	✓	✓	✗	✗	93.01	93.87	92.45
04	✓	✓	✓	✓	✗	93.69	94.20	93.18
05	✓	✓	✓	✗	✓	93.31	93.75	92.77
06	3D+2D	✓	SA	✗	✓	90.01	90.89	89.20
07	✓	✓	✗	✓	✓	91.29	92.10	90.59
08	3D+2D	✓	✓	✓	✓	94.11	94.05	93.89
09	✓	✓	✓	✓	✓	**94.81**	**94.73**	**94.39**

E. Ablation Analysis

1) Ablation Analysis of Different Modal Data Inputs: To more intuitively demonstrate the classification advantages of HSIs combined with LiDAR elevation information, we conducted ablation experiments on different datasets targeting the input data sources. In the first set of experiments, we used only HSIs as the input to the model on different datasets. In the second set of experiments, we used only LiDAR data as input for the model on different datasets. As shown in Table VII, using multisource data as input achieved the best classification results, further validating the effectiveness of the proposed model in fusing multisource data.

2) Ablation Analysis of Different Components: To verify the effectiveness of each component of the proposed model, we performed ablation experiments on these components. Specifically, the model mainly consists of the following modules: *MMC*, *FTT*, *CCA*, *Mambaunit*, and *MLP*.

As shown in Table VIII, we proposed eight variant schemes on the Houston dataset. The first scheme retained only the *MMC* component, and we found that the classification performance was significantly reduced. The second scheme retained both the *MMC* and *FTT* components, and this variant improved the OA by 13.83% compared to the previous variant. The third scheme removed the *Mambaunit* and *MLP* components, and we found that this variant reduced the OA by 1.8% compared to MCCGM, demonstrating the effectiveness of *CCA* in capturing long-sequence dependencies and fusing multisource data. The fourth scheme removed the *MLP* component, and due to *MLP*'s ability to enhance the model's representation capability and local information processing through nonlinear transformation and feature dimension expansion, this variant reduced the OA by 1.12% compared to MCCGM. The fifth scheme removed the *Mambaunit*, resulting in the model losing some morphological feature capture capability, and this variant reduced the OA by 1.5% compared to MCCGM. The sixth scheme replaced the *MMC* component with a simple combination of 3-D and 2-D convolutional layers (3-D + 2-D), replaced *CCA* with standard SA, and removed the Mamba unit. We found that this variant reduced the OA by 4.8% compared to MCCGM. The seventh scheme removes the *CCA* module and uses simple element-wise addition to input the result into the *Mambaunit*. We found that this variant reduced the OA by 3.71% compared to MCCGM. The last scheme replaced the *MMC* component with a simple combination of 3-D and 2-D convolutional layers (3-D + 2-D), while keeping the other components unchanged. Compared with MCCGM, this variant reduces OA by 0.74%. This indicates that *MMC* effectively extracts multiscale morphological features of different modalities, which can effectively serve the classification results. This analysis further confirms the rationality and effectiveness of the proposed model and its components.

IV. CONCLUSION

This article proposes a joint HSI–LiDAR classification model that integrates multiscale convolution with an MFE. A three-branch network framework is designed to achieve effective feature extraction. The *MMC* module extracts multiscale spatial–spectral features from HSI and elevation features from LiDAR, while also capturing structural information through morphological convolution. The proposed *CCA* module further exploits the complementary nature of multimodal data and mitigates redundant feature generation. Finally, the MFE models sequence dependencies and mine deeper morphological representations, thereby enhancing the accuracy of joint classification. Extensive experiments on four public datasets demonstrate that MCCGM achieves superior classification performance and strong generalization compared with state-of-the-art methods. Overall, the proposed MCCGM effectively fuses multisource data and delivers robust classification results.

The HSI provides rich spectral information for land-cover differentiation, while LiDAR offers complementary elevation and structural cues that are especially useful for distinguishing classes with similar spectral responses. Their joint use can effectively alleviate the limitations of single-modality classification and improve robustness in complex scenes. Therefore, multisource fusion is not only beneficial for enhancing classification accuracy but also important for fine-grained and reliable remote sensing interpretation. In the future, this direction is expected to become increasingly important in large-scale land-cover mapping, complex scene understanding, and intelligent Earth observation applications. In future work, we will

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investigate methods for reducing redundant features, developing more efficient morphological operations, and achieving adaptive multimodal feature fusion to build a joint classification model that offers higher accuracy and improved computational efficiency.

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